

PANCE Histology & Microscopy Findings

Buzzword findings that appear in PANCE clinical vignettes (grouped by organ system)

Note: Histology is not a formal PANCE task category. The 2025 NCCPA blueprint tests pathophysiology under *Applying Foundational Scientific Concepts (8%)*, with the skill of *relating pathophysiologic principles to specific disease processes*. Microscopy/biopsy/cytology results appear inside clinical vignettes - this chart covers the high-yield buzzwords PA prep resources consistently emphasize.

Finding	Associated Disease	Why It Matters Clinically
HEMATOLOGIC / LYMPHOID		
Reed-Sternberg cells	Hodgkin lymphoma	Bimodal age distribution; staging with PET-CT; curative-intent ABVD chemo
Auer rods	Acute myeloid leukemia (AML)	If APL/M3 with t(15;17): treat with ATRA + arsenic (DIC risk before treatment)
Smudge cells (peripheral smear)	Chronic lymphocytic leukemia (CLL)	Often incidental on routine CBC in older adults
Howell-Jolly bodies	Asplenia / post-splenectomy	Mandates pneumococcal, meningococcal, Hib vaccination
Heinz bodies / bite cells	G6PD deficiency hemolysis	Avoid sulfa drugs, fava beans, primaquine, dapsone
Hypersegmented neutrophils	B12 / folate deficiency (megaloblastic anemia)	Distinguishes megaloblastic from iron-deficiency microcytic anemia
Schistocytes (helmet cells, fragments)	MAHA - TTP, HUS, DIC, HELLP	TTP pentad = emergency; plasma exchange; do NOT transfuse platelets in TTP
Spherocytes	Hereditary spherocytosis OR warm autoimmune hemolytic anemia	Direct Coombs: positive = AIHA; negative = HS (osmotic fragility +)
Sickle cells / target cells / Howell-Jolly	Sickle cell disease	Confirm with Hgb electrophoresis; vaso-occlusive crisis = pain + hydration + opioids
Tear-drop cells (dacrocytes)	Myelofibrosis (or other marrow infiltration)	Often with leukoerythroblastic smear and dry tap on bone marrow aspirate
Rouleaux formation	Multiple myeloma	Pair with M-spike on SPEP, hypercalcemia, lytic bone lesions, renal failure
Basophilic stippling	Lead poisoning, sideroblastic anemia, thalassemia	Lead in children → chelation; check Pb, FEP/ZPP
Target cells	Thalassemia, HbC, liver disease, asplenia	Helps narrow microcytic anemia differential
MSK / JOINT ASPIRATE (POLARIZED MICROSCOPY)		
Needle-shaped, negatively birefringent crystals (yellow parallel)	Gout (monosodium urate)	Acute: NSAIDs, colchicine, or steroids. Chronic: allopurinol/febuxostat (NOT during flare)
Rhomboid, positively birefringent crystals (blue parallel)	Pseudogout (CPPD)	Knees most common; chondrocalcinosis on X-ray; NSAIDs/colchicine/steroids
Apple-green birefringence on Congo red stain	Amyloidosis (AL, AA)	Suspect with nephrotic syndrome + restrictive cardiomyopathy + neuropathy
PULMONARY		
Non-caseating granulomas	Sarcoidosis	African-American adult, bilateral hilar adenopathy; ACE may be elevated; treat with steroids if symptomatic
Caseating granulomas (Ghon focus / complex)	Tuberculosis	RIPE therapy (rifampin, isoniazid, pyrazinamide, ethambutol) + contact tracing
Ferruginous bodies	Asbestosis	Pleural plaques on CXR; ↑ risk of mesothelioma and bronchogenic carcinoma
Curschmann spirals / Charcot-Leyden crystals	Asthma (sputum)	Eosinophilic inflammation; supports asthma over COPD when smoker is unclear
ENDOCRINE		
Orphan-Annie eye nuclei + psammoma bodies	Papillary thyroid carcinoma	Best-prognosis thyroid cancer; total thyroidectomy ± radioactive iodine
Amyloid stroma (calcitonin staining)	Medullary thyroid carcinoma	Screen for MEN2 / RET mutation; check calcitonin and CEA
Lymphocytic infiltrate with germinal centers (Hurthle cells)	Hashimoto thyroiditis	Anti-TPO and anti-thyroglobulin antibodies; treat with levothyroxine
GASTROINTESTINAL / HEPATIC		

Finding	Associated Disease	Why It Matters Clinically
Non-caseating granulomas + skip lesions (terminal ileum)	Crohn disease	Steroids/biologics; surgery is NOT curative (transmural, mouth-to-anus)
Crypt abscesses, continuous mucosal inflammation	Ulcerative colitis	Limited to colon; total colectomy is curative; ↑ risk of colon cancer
Signet-ring cells	Diffuse-type gastric adenocarcinoma (linitis plastica)	Aggressive; H. pylori is the major risk for the intestinal type
RENAL / URINE SEDIMENT		
RBC casts / dysmorphic RBCs	Glomerulonephritis (nephritic syndrome)	Hematuria + HTN + edema + proteinuria; identifies glomerular bleeding
WBC casts	Pyelonephritis or acute interstitial nephritis (AIN)	Pyelo: fever + flank pain + CVA tenderness. AIN: drug reaction (NSAIDs, PPIs, abx)
Muddy-brown / granular casts	Acute tubular necrosis (ATN)	Most common cause of intrinsic AKI; ischemic or nephrotoxic
Broad / waxy casts	Chronic kidney disease	Indicates chronic tubular dilation
Fatty casts / oval fat bodies (Maltese cross)	Nephrotic syndrome	Heavy proteinuria (>3.5 g/day), hypoalbuminemia, edema, hyperlipidemia
REPRODUCTIVE / VAGINITIS		
Koilocytes (Pap smear)	HPV-related cervical dysplasia	Triggers colposcopy per ASCCP guidelines
Clue cells (saline wet mount)	Bacterial vaginosis	Whiff test +, pH >4.5; treat with metronidazole
Pseudohyphae / budding yeast (KOH)	Vulvovaginal candidiasis	Treat with fluconazole 150 mg or topical azole
Motile trichomonads (saline)	Trichomoniasis (STI)	Strawberry cervix; treat patient AND partner with metronidazole
INFECTIOUS DISEASE		
Owl's-eye intranuclear inclusions	Cytomegalovirus (CMV)	Especially in HIV (CD4 <50: retinitis) and transplant; treat with ganciclovir
Multinucleated giant cells (Tzanck smear)	HSV or VZV	Distinguishes herpetic vesicles from other vesiculobullous conditions
Gram-positive lancet-shaped diplococci	Streptococcus pneumoniae	Most common cause of CAP and bacterial meningitis in adults
Acid-fast bacilli (Ziehl-Neelsen)	Mycobacterium tuberculosis (or NTM)	Confirm with NAAT/culture; report to public health
Gram-negative diplococci (intracellular)	Neisseria gonorrhoeae	Cervical/urethral discharge; ceftriaxone + doxycycline (chlamydia coverage)
DERMATOLOGY		
Acantholysis / tombstone basal cells (Tzanck)	Pemphigus vulgaris	IgG vs desmoglein-3; Nikolsky sign +; mortality if untreated; steroids
Subepidermal bulla with eosinophils	Bullous pemphigoid	IgG vs hemidesmosomes; tense bullae; Nikolsky negative; better prognosis than PV
Munro microabscesses (parakeratosis, no granular layer)	Psoriasis	Usually clinical dx; biopsy if uncertain; treat with topical steroids, vit D analogs, phototherapy, biologics
NEURO		
Lewy bodies (α-synuclein inclusions)	Parkinson disease / Dementia with Lewy bodies	DLB triad: fluctuating cognition, visual hallucinations, parkinsonism. AVOID typical antipsychotics